

Scientific review of intentional content retrieval for digital wellbeing

Critical literature review and proposed research program

Evidence reviewed through 28 September 2026

Assessment at a glance

The concept is scientifically plausible and worth testing, but its complete causal claim remains unproven. Experiments show that particular interface features and access constraints can change engagement; small HCI studies show that useful functions can sometimes be retained while distracting features are restricted. Neither result establishes that a general content-extraction intermediary reduces unwanted use, preserves most valued utility, and avoids displacement over months. [1, 2, 3, 4, 5]

The most defensible initial claim is narrower: **for users with a specific retrievable information need, an accurately scoped interface may reduce exposure to opportunities for unwanted continuation.** This is a proposed product hypothesis. Evidence is strongest for the manipulability of engagement and weakest for comprehensive utility preservation, durable reductions in compulsive behavior, and superiority to a well-configured blocker. [6, 7, 8, 2]

Two findings materially constrain the design. First, opening a phone can reflect boredom, connection-seeking or habit rather than a missing fact, so supplying information may leave the underlying motive intact. Second, wellbeing includes valued connection and autonomy, requiring outcomes beyond minutes. [9, 10, 11]

The proposed development gate is therefore a **joint result**: less unwanted engagement across the measured digital ecosystem, together with preserved task success and valued social functionality. A nonsignificant difference in utility is insufficient to establish preservation; use an explicit noninferiority margin. [12]

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Scope and method

This is a structured critical review, not a preregistered systematic review or a new meta-analysis. Searches covered web-indexed publisher records, PubMed/PMC, ACM proceedings and author or institutional copies, supplemented by references in relevant systematic reviews. Search families combined smartphone/social-media problems, automaticity/checking, engagement features, interventions, information retrieval/filtering, named HCI systems, measurement and displacement. Dedicated searches sought 2025-2026 publications, null results, adverse effects and corrections. The bibliography includes verified material available by the review date; it does not imply coverage of the rest of 2026.

Priority went to systematic reviews, randomized field studies, objective behavioral measurements and original HCI system evaluations. Conceptual work is labeled as such. A small usability study is useful evidence of feasibility, but not equivalent to a randomized trial of sustained behavior. Public clinical sources establish terminology rather than product efficacy. Where the accessible primary record did not permit verification of a sample size or demographic detail, the report says **NR** rather than supplying a guess. Direct publisher/DOI links and accessible original manuscripts are provided in the bibliography and evidence tables.

Important limitations are English-language and web-access bias, incomplete full-text access, selective indexing, overlapping cohorts and reviews, and absence of formal duplicate screening or a registered search protocol. No exhaustive search-flow counts are claimed. Negative search findings mean that qualifying evidence was not located in this review, not that none can exist. The research should be updated before a confirmatory study is preregistered.

How evidence is graded

Grade	Meaning in this report
Strong	Convergent credible evidence or a large direct experiment supports the precisely stated outcome. It does not automatically generalize to clinical impairment or this product.
Moderate	Credible causal or convergent longitudinal evidence, with material limitations in replication, precision, setting or measurement.
Preliminary	Small exploratory experiments, prototype deployments, observational studies or indirect extrapolation.
Mixed	Credible findings differ by study, outcome, population or implementation.
Unsupported / insufficient evidence	Direct evidence was not located, or available work cannot support the proposed causal claim. This is not proof of no effect.

These are review judgments, not formal GRADE ratings. Separately, **A** denotes experimental evidence; **B**, longitudinal/repeated observational evidence; **C**, cross-sectional/correlational evidence; **D**, theory; and **E**, popular claims not adequately supported here. Randomization can support causality for the measured outcome while evidence for a broader claim remains preliminary.

Reading the evidence tables

Seven requested evidence tables appear after the research program. They distinguish actual telemetry from device values manually supplied by participants, objective laboratory tasks, and self-report. The numbered bibliography supplies full titles, authors, venues, DOI links and study details. Repeated entries across tables identify the same study, not independent replications. Journal issue years are used, with earlier online/preprint dates noted where relevant.

1 Problematic smartphone and social media use

Distinguish quantity automaticity control and harm

Term	Operational meaning	What it does not establish
Heavy use	High frequency or duration relative to a defined comparison group	Loss of control, distress or impairment. Intense use and problematic use show different associations. [13]
Habitual use	A learned tendency to act automatically in recurring contexts	That the behavior is unwanted or harmful. Frequency and automaticity are related but distinct. [14, 15]
Problematic use	Dysregulated use associated with adverse consequences or functional interference	A universally agreed clinical diagnosis or a particular number of hours. [16, 17]
Compulsive use	Repeated difficulty resisting or stopping despite an intention to do so, often despite costs	A diagnosis inferred from rapid swiping or frequent checking alone. [16, 18]
Addiction	A clinical or theoretical construct requiring an explicit framework and meaningful impairment	An automatic interpretation of a questionnaire title or screening cutoff. [18, 19]

This is a vocabulary for measurement, not a five-step severity ladder. A clinician working on a phone can be a heavy user; an automatic calendar check can be useful; occasional unplanned leisure can be valued. Context and the user's goals belong in the outcome definition. The distinction is consistent with a conception of digital wellbeing that includes autonomy, connection and benefits, rather than only absence of symptoms. [11, 9]

The American Psychiatric Association's current public guidance does not identify social-media addiction or the technology addictions it discusses as DSM-5-TR diagnoses; Internet Gaming Disorder remains a condition for further research. WHO's ICD-11 gaming disorder is specifically about gaming and requires significant impairment, normally evident over at least 12 months. These gaming classifications should not be generalized to all smartphone use. [18, 19]

Behavioral-addiction frameworks can organize hypotheses about impaired control, cue-reactivity, craving, reinforcement and persistence despite costs. They do not prove that a phone is the addictive object or that all high-engagement behavior shares one mechanism. The I-PACE model is a process framework linking person characteristics, affect/cognition and control; the pathways account distinguishes different routes to problematic phone behavior. Both are theoretical explanations to test. [20, 21]

The counterargument is substantive: importing symptom checklists from substance use or gaming can pathologize common behavior without demonstrating independent impairment. A 2025 adolescent psychometric analysis distinguished core problems from salience/tolerance, while a 2026 Swedish study added nuance rather than validating one universal symptom structure. The report therefore favors continuous difficulties and concrete interference over declaring participants “addicted.” [22, 17, 23]

Mechanisms with different evidential status

Habit and informational reward. Oulasvirta and colleagues described brief checking routines that sometimes opened a route into other applications; a small field manipulation showed that rapidly available dynamic information could reinforce checking. This supports a gateway mechanism, while its early devices and tiny intervention groups limit direct extrapolation to modern feeds. [15]

Social reward and reinforcement learning. Lindström and colleagues combined more than one million posts from over 4,000 users with an experiment of 176 participants. Manipulated social rewards changed posting behavior in a way consistent with reward-learning models. This is behavioral evidence about reward-sensitive action, especially posting; it does not measure dopamine release or establish an addiction mechanism for every scroll. [24]

Boredom and affect regulation. Seven experiments found a feedback pattern in which boredom encouraged digital switching and switching often increased boredom; generalization was less consistent for articles and nonstudent samples. Observed phone use also includes breaks and habitual checking that users do not evaluate negatively. Extraction may solve the wrong problem when the goal is stimulation, escape or companionship. [10, 9]

FOMO and vigilance. Fear of missing rewarding experiences is associated with engagement; online vigilance separates mental salience, responsiveness and monitoring. These are plausible sources of recurrent checking, but trait correlations do not establish that feeds cause FOMO or that supplying a digest will relieve it. Vigilance can accompany both stress and felt connection. [25, 26]

Cue-reactivity and craving. A small extreme-group study found greater reported craving to social-media cues, but nonclinical classification and fixed cue order constrain causal interpretation. [27] Treat cue-evoked attention, a reported urge and persistent impaired control as separate endpoints. A notification can interrupt performance without anyone opening the phone, and a craving questionnaire can capture subjective urge without detecting an app-entry event. Evidence for a cue effect is consequently not evidence that all subsequent use is compelled. [28, 29]

Contextual repetition. A 2025 sensing study associated stronger phone habits with familiar spatial routines. This supports studying context and repeated situations, while its observational design cannot show that changing location would reduce use. [30]

Conclusion: Moderate evidence that habit, reward and contextual cues contribute to engagement; **Mixed** evidence that these mechanisms necessarily produce harm; **Unsupported** inference that a single general smartphone-addiction mechanism explains all cases. [15, 24, 31, 9, 20]

2 Why users stay longer than intended

The proposed sequence is credible, but its stages are rarely measured together. The most relevant work includes checking-to-other-app transitions, feature-level regret, reported absorption, and experience-sampling links between automaticity and task delay. These studies support the existence of intention drift, not an established population percentage of app openings that become unwanted consumption. [15, 32, 33, 34]

Baughan and colleagues deployed a modified Twitter client for four weeks with 43 users. Participants described both restorative absorption and experiences of losing time with poor recall; interface changes reduced reported dissociation. The construct was everyday absorption, not a diagnosis of dissociative disorder, and the intervention bundled features. [33]

In a preregistered study of 1,315 adults, with logs available for 691 Android users, mindless scrolling was associated with guilt and very small daily wellbeing differences. Crucially, the hybrid measure multiplied reported goal-less use by logged time: the phone did not objectively detect awareness, scrolling or intent. This is close to the product's construct but remains observational. [35]

Adolescent experience sampling linked automaticity to delayed tasks on average, with substantial person-specific variation. A separate student diary study found small or absent direct habit-wellbeing relationships. These results favor measuring actual goal interference rather than assuming every habit is harmful. [34, 31]

Two adjacent literatures strengthen particular links. Notification experiments demonstrate immediate distraction without responding, and a small short-video experiment tests prospective memory after interruptions. General attention-residue experiments concern carrying attention from one work task into the next. None independently proves that a user forgot the reason for opening a specific social app, or that an extraction intermediary prevents this. [28, 36, 37]

For product research, distinguish **entry without a specific intention**, **intentional entry followed by a useful new goal**, **intentional entry followed by unwanted continuation**, and **deliberately chosen open-ended recreation**. Collapsing them into “unplanned use” would misclassify beneficial discovery and make utility preservation look easier than it is. Feature-level research demonstrates why whole-app good/bad labels are too coarse. [32]

Conclusion: **Moderate** evidence that interface exposure can alter continuation and that drift occurs; **Preliminary** evidence for the complete need-to-entry-to-cue-to-unwanted-use mediation chain; **Insufficient** direct evidence for its prevention by a general extraction product. [1, 4, 15, 5]

3 Engagement and persuasive design mechanisms

The table grades the precise outcome supported. It is intentionally more cautious about compulsive impairment than about clicks or minutes.

Mechanism	Evidence type	What the evidence supports	What remains unproven
Algorithmic and personalized feeds	A field experiments	Chronological ranking reduced logged engagement on Meta platforms; 2026 X findings were asymmetric and survey-based. [6, 38]	Universal effects across algorithms, improved wellbeing, or prevention of compulsivity. A 2026 custom-feed trial found no significant engagement-behavior difference. [39]

Mechanism	Evidence type	What the evidence supports	What remains unproven
Infinite scroll and endless availability	A small interface experiments; B deployments; D	A 2026 TikTok prototype isolates scroll friction; stopping prompts have context-dependent responses. [40, 41]	Durable benefit of pagination alone, or a general harmful duration threshold.
Autoplay	A field and laboratory experiments	A Netflix experiment changed logged watching behavior; TikTok feature changes altered interaction proxies. [4, 40]	Equal effects on every platform; reduced clinical impairment or improved agency.
Push notifications	A laboratory and field; B/C	Alerts can impair immediate attention. Disabling all alerts has mixed behavioral effects. [28, 42]	That all checking is notification-driven or that silence reliably reduces time.
Badges	A narrow interface experiment	Badge placement changed first-click choice in a static-screen experiment. [43]	Effects of badge color specifically, chronic compulsion, or durable benefits from removal.
Likes and social feedback	A reward experiment and fMRI task; C traces	Social rewards influence posting; peer approval changes adolescent responses in a simulated task. [24, 44]	Direct dopamine release, universal loss of control, or that hiding counts reduces problematic use.
Comments and follower counts	C qualitative; D; limited component A	A TikTok social-investment obstruction had negligible effects. [40]	Separate durable behavioral effects of removing comments or follower counts.
Variable and intermittent rewards	A reward-rate evidence; D extrapolation	Social reward learning is supported. [24]	A demonstrated variable-ratio schedule for every feed or a universal slot-machine equivalence.
Pull-to-refresh	D; E when asserted as settled addiction mechanism	Plausible action-to-novel-content loop, related indirectly to informational checking. [15]	A qualifying isolated longitudinal/RCT demonstration that the gesture itself causes compulsive use was not located.
Short-form video and rapid switching	A small lab experiments; restricted allocation in the prospective-memory study	Certain interruptions impair prospective memory; switching can increase boredom. [36, 10]	Permanent attention damage, a general diagnosis, or equivalent effects of duration, video length, ranking and swipe mechanics.
Engagement optimization and persuasive design	A for tested rankings/features; D for broad theories	Changing objectives or interface options can alter exposure and behavior. [6, 39]	That every persuasive feature is harmful, deceptive, or causally equivalent.

The Netflix autoplay experiment involved 76 users. Its reported reductions were relative to control and baseline, while both groups increased daily watching from baseline. This is stronger evidence for a specific feature than a generic claim that users are manipulated, but its short duration and analysis warrant replication. [4]

The 2026 TikTok experiment involved 48 participants and found tradeoffs: changes to autoplay/scrolling reduced engagement proxies but worsened experience and did not reliably increase agency. “Compulsive engagement patterns” in that paper must not be read as diagnosed compulsivity. [40]

Neural activation in a reward-related region during a likes task is not a direct measurement of dopamine and cannot establish loss of control. Likewise, a fitted reinforcement-learning model does not establish a neurotransmitter mechanism. The claim “social media hijacks dopamine” is too

imprecise for this review and unsupported as a product rationale. [44, 24]

The design implication is to test particular features with matched content and measured outcomes. Labels such as persuasive design or dark patterns can describe a design concern, but they cannot substitute for isolating the feature and its effect. [45, 40]

4 Interventions that reduce problematic use

What systematic reviews establish

The 2023 digital self-control review covered 62 articles and reported a small-to-medium pooled reduction in time. Its pooled estimate largely compared intervention periods with baseline, including within intervention arms, so it should not be presented as a pooled randomized treatment effect. Studies were usually short and often small. [7]

A review of 21 detox studies found positive, null and negative outcomes. A 2025 meta-analysis of 10 abstinence studies involving 4,674 participants found no significant average effects on positive affect, negative affect or life satisfaction; objective adherence measurement was uncommon. These findings challenge universal detox claims, without proving that no individual benefits. [46, 47]

A 2026 umbrella review of internet, smartphone and gaming interventions reported symptom improvements but generally low certainty, substantial heterogeneity and overlap among reviews. CBT, exercise or multimodal results from these mixed populations do not establish that a consumer information intermediary treats an addiction. [48]

Behavior change with comparatively credible evidence

App limits and commitment. The Digital Addiction field experiment used objective Android logs and randomized limits/incentives. Limits reduced target-app use by about 22 minutes/day over 12 weeks, but approximately 12 minutes shifted to other apps. The study's economic habit/self-control model and title are not psychiatric diagnoses. [3]

Selective access removal. The 2025 mobile-internet blocking trial retained calls and texts and found improvements in attention-task performance and reported wellbeing. However, only 119 of 467 randomized participants met the specified adherence criterion, and the primary analyses used complete survey cases. This supports further study of restricted access while limiting confidence in broad real-world uptake. [49]

Time reduction. A randomized trial in 111 healthy students found short-term mental-health and sleep benefits from a three-week screen-time reduction regimen, followed by rapid rebound in use. It did not measure whether desired information or relationships were preserved. [50]

Grayscale and settings bundles. A randomized field study of 112 iPhone users compared grayscale, self-set limits and monitoring; grayscale reduced time immediately and limits more gradually, without immediate wellbeing or academic gains. A separate nudge-bundle RCT improved some outcomes, but cannot identify which of ten strategies caused the effect. [51, 52]

App-entry friction. The one sec paper combined a 280-user uncontrolled field cohort with a 500-person randomized online component experiment. Its widely repeated 57% reduction in completed opens is a before-after field result, not a randomized field effect. The experiment supports a dismiss option and delay more clearly than deliberation wording. Longer observational follow-up cannot remove selection and survivorship bias. [53, 54]

Notification suppression batching and previews

Fitz and colleagues randomized 237 Android users to usual delivery, hourly batches, three batches/day or no notifications. Three daily batches improved several self-reported outcomes; the unlock result was suggestive, with many unadjusted comparisons and a nonsignificant omnibus test. No-alert delivery did not reliably reduce unlocks and increased FOMO/anxiety. [55]

A later preregistered notification-disabling RCT analyzed 205 of 274 randomized young adults. It found no significant effect on logged checking or screen time, while situational FOMO increased. Post-randomization exclusions and a one-week treatment limit interpretation. [42]

Other field experiments report less perceived distraction or stress under reduced interruptions or scheduled email checking, but rely largely on self-report and sometimes bundle phone placement with alerts. Observational silent-mode findings cannot show that muting causes more checking. [56, 57, 58, 59]

Thus notification batching is a promising configurable intervention, not a proven universal schedule. Standalone preview or digest extraction requires direct testing of retrieval success, checking and missed information.

Intentional use adaptive intervention and psychological approaches

A goal-directed training RCT compared Not Less But Better with daily detox plans. Both groups improved on some measures, but intentional-use training was not superior and unlock frequency did not change; substantial attrition complicates interpretation. This directly weakens the assumption that a more intentional framing must outperform a simpler restriction. [60]

MyTime selectively targeted apps users considered poor uses of time and reported a 21% reduction without changing time in good-use apps. This is encouraging but uncontrolled, and unchanged good-app minutes are not equivalent to preserved information-retrieval success. [61]

Adaptive systems such as Time2Stop and MindShift provide early field evidence that context-sensitive intervention delivery can alter app visits or duration. They are behavior-change prompts, not evaluations of AI content extraction. A 2026 Wellspent RCT reduced target-app time but did not significantly improve problematic social-media-use scores or self-efficacy. [62, 63, 64]

Mindfulness combined with self-monitoring and mood tracking improved reported distraction in a short student RCT but did not reduce habitual behavior or social-media time. CBT-based approaches have broader symptom-oriented evidence, whose certainty and relevance to this product remain limited. Neither should be described as a proven software component for preserving utility. [65, 48]

Coverage of the remaining intervention categories

Intervention family	Current assessment for this concept
Screen Time and Digital Wellbeing dashboards, limits, goal-setting	Some device-logged benefit for configured strategies; monitoring alone is an imperfect active control and may change behavior. A product feature's availability does not establish efficacy. [51, 60]
Notifications disabled or batched	Mixed; detailed above. Batching is promising, universal silence is not reliably effective. [55, 42]
App removal, blockers and temporary abstinence	Can constrain opportunity; wellbeing and adherence vary. Broad disconnection trials do not isolate deletion from access loss. [46, 49]
Feed blockers, recommendation removal and minimalist interfaces	Promising feature-specific behavior evidence; small and bundled studies dominate. [1, 66, 2]
Dumbphones, minimalist phones and launchers	No strong directly comparable controlled evidence located for sustained unwanted-use reduction with preserved utility. Mobile-internet blocking is an analogy, not a validation of hardware or launcher products. [49]
Delayed access, friction and nudges	Causal evidence for proximal decisions; long-term independent effects and utility remain less established. [53, 54]
Self-monitoring, values and goal-setting	Frequently bundled; evidence for unique component superiority is mixed. [7, 60]
CBT and mindfulness	Some symptom/perceived-distraction benefits; distinct from objective ecosystem engagement and this product's utility claim. [48, 65]
Just-in-time adaptive interventions	Early field prototypes plus mature trial-design methodology; automatic adaptation is not itself proof of efficacy. [62, 67, 68]

Conclusion: Moderate evidence that several strategies can reduce measured behavior in selected users, **Mixed** evidence for lasting wellbeing/impairment gains, and **Insufficient** evidence that a single strategy is best for everyone. [7, 47, 50, 51, 60]

5 Preserving utility while reducing engagement

The relevant unit is often a feature or activity, not an entire app. Feature-level studies find different rewards and regret within the same service. Removing a recommendation panel while retaining a specific video, conversation or community is therefore a coherent target, but the distinction must be tested with the person's actual goals. [32, 2]

A small randomized Facebook study compared goal reminders, newsfeed removal and a visual control. The interventions reduced several logged engagement outcomes, while interviews identified both better focus and lost information/FOMO. It demonstrates selective interface modification, but not full preservation of utility or superiority to blocking. [1]

FinerMe compared app-level and feature-level feedback and restriction in a 16-day study of 56 users. Use declined during restriction in both groups; feature controls better targeted some passive features, without a clear overall-time advantage. Allocation was not verified as random, and the study lacked an untreated control. [2]

Purpose Mode allowed users to disable attention-capturing features across several websites. Its 29-person before-after deployment reported less distraction and lower logged website time; demand,

novelty, fixed period order and unmeasured phone use prevent a clean causal interpretation. [66]

Utility loss is an observed risk, not only a hypothetical objection. In a four-week randomized Facebook deactivation study, the impact-evaluation sample comprised 1661 US users. Subjective wellbeing improved, but factual news knowledge declined; replacement activities included both socializing and television. Compliance checks and the news quiz were objective, whereas most time-use measures were self-reported. Selective incentive eligibility and the election setting limit generalization. [69]

Taken together, these studies justify a **selective-functionality prototype** and a utility evaluation. They do not show that extraction preserves “most” utility: none establishes a population-wide denominator including interaction, serendipity, entertainment, emotional support, participation and messages requiring replies. “Most” must be defined before a study, by task category and user importance. [8, 2, 1, 32]

Answer to the requested comparison: No convincing direct extraction-versus-blocking trial establishing both better utility and comparable unwanted-use reduction was located. FinerMe is a close component-level comparison, while SwitchTube demonstrates agency tradeoffs. The key differentiation claim remains **Unsupported / insufficient evidence**, rather than refuted. [2, 8]

6 Research most analogous to the proposed product

The concept has important predecessors. Their proximity should be judged on three separate dimensions: expressing an intention, obtaining content through another interface, and demonstrating a wellbeing benefit.

System or research line	Why it is analogous	What its evaluation establishes
Bonsai, CHI 2026	Natural-language intent drives source selection, curation and ranking outside an ordinary default feed.	Fifteen Bluesky users could build intentional feeds; effort and transparency mattered. It remains a feed and does not establish reduced compulsive use. [5]
SwitchTube, CHI 2023	Alternative YouTube interface offers search-first Focus, recommendation-first Explore and a switch between modes.	See Section 7 for the field comparison and its limitations. [8]
FinerMe, CSCW 2023	Restricts selected features while retaining other functions.	See Section 5 for the comparative evaluation. [2]
Purpose Mode, TOCHI 2025	Removes selected distraction features across source websites.	Promising but uncontrolled logged behavior changes. [66]
TwitInfo, CHI 2011	An event query retrieves microblog content into a summarized event timeline.	A concrete query-to-social-content antecedent; usability and event extraction, not compulsivity outcomes. [70]
FeedWinnower, CHI 2010	Filters aggregated streams by topic, person, source and time.	Small formative evaluation of narrowing information streams; not a wellbeing trial. [71]
Reader View, CHI 2019	Extracts article content from the original webpage presentation.	Reading-speed and readability improvements; occasional essential-content loss; no reduced compulsive-use test. [72]

System or research line	Why it is analogous	What its evaluation establishes
HabitLab, CSCW 2018 and CHI 2019	Delivers interface changes, including hiding feed-like elements, with cross-site logging.	Behavioral change and adherence/substitution evidence; not intent-to-answer extraction. [73, 74]
Notification digests and scheduled email checking	Deliver or inspect information at bounded times without reacting to every arrival.	Some attention/stress benefits; no proof that arbitrary previews satisfy needs without inducing more checking. [55, 58]

RSS, aggregators, read-it-later tools, reader modes and personal information agents should be treated as architecture families, not interchangeable treatments. FeedWinnow and TwitInfo establish useful technical antecedents. This review did not locate strong controlled evidence that ordinary RSS or read-it-later adoption, notification previews alone, or a general assistant reduces compulsive smartphone use while preserving utility. [71, 70, 72]

AI summarization requires an additional distinction: compression can change meaning. A 2026 experiment with 144 participants found that social-discussion summaries affected opinion formation, without changing willingness to engage. Earlier summarization benchmarks demonstrate that fluent text need not be faithful; those historical error rates cannot be transferred to current models. [75, 76]

Conclusion: **Preliminary** direct HCI support for the interface family; **Insufficient** evidence for the proposed combined behavioral and utility outcome. [5, 8, 2]

7 Search versus feed interaction

Search externalizes an information request, which makes it attractive for deliberate retrieval. However, search can also be exploratory: a person may learn, investigate and reformulate goals rather than complete a single lookup. Query input alone is therefore not a scientifically established stopping mechanism. [77]

YouTube users report that search and playlists support agency more than autoplay and recommendations, but these are subjective feature attributions. SwitchTube provides a more direct comparison: search-first Focus produced higher perceived agency and satisfaction, while its descriptive median use exceeded Explore. The modes bundled interface differences; Focus/Explore order was counterbalanced, while Switch always came last. [78, 8]

Chronological-versus-algorithmic studies answer a different question. The large Meta experiment reduced logged use with chronological feeds, without detectable changes in its principal political outcomes. The 2026 X experiment found increased survey-based engagement when activating the algorithm but no significant reversal when disabling it. Neither compares a query-only interface with a feed. [6, 38]

A 2026 registered-report experiment on custom feeds found that ranking changed content and social-norm perceptions without significantly changing measured engagement behavior. Thus a recommender's objective and corpus matter: replacing engagement optimization can change quality without simply reducing quantity. [39]

Assessment: **Preliminary** evidence that search-first interfaces support perceived intentionality for specific tasks; **Mixed/insufficient** evidence that they reduce time or unwanted use; **Unsupported** claim that all query interfaces are inherently bounded or all recommendation interfaces undermine wellbeing. [8, 77, 39]

The clean experiment crosses query versus feed presentation with finite versus continuously available content. Match relevance, corpus, latency and task, then measure utility and unwanted continuation separately. If stopping points explain the effect, that supports bounded presentation rather than a unique benefit of asking a question. This is a proposed causal test.

8 Measurement

Validated instruments and their limits

Construct	Instrument	Validation and recommended interpretation
Smartphone problem severity	SAS-SV, 10 items	Original N=540 Korean adolescents. Internal consistency was high; sex-specific screening thresholds are not universal diagnostic cutoffs. Use continuous scores and functional outcomes. [79]
Social-media problem severity	SMD, 9 items; longer 27-item form	Developed in 2,198 Dutch adolescents. Past-year reference is poorly suited to detecting week-to-week change without an explicitly validated adaptation. [80]
Social-media symptoms	BSMAS, 6 items	Introduced in a large Norwegian survey. Core versus peripheral symptom structure is disputed; salience/tolerance need careful interpretation. [81, 17, 23]
Broad habit strength	SRHI, 12 items	Includes repetition, identity and automaticity; this breadth can blur the intended mechanism. [82]
Automaticity	SRBAI, 4 items	Behavior-specific automaticity index with substantial validation across mainly health/travel behaviors; phone adaptations should be tested. [14]
FOMO	FoMOs, 10 items	Trait fear of missing rewarding experiences, developed across three studies; not momentary craving or a diagnosis. [25]
Online vigilance	OVS, 12 items	Salience, reactivity and monitoring; four German validation samples. It captures persistent cognitive connection, which is not inherently pathological. [26]
Craving	Social Media Craving Scale, 5 items	Turkish university validation N=423; limited generalizability and some weak fit in a secondary sample. Distinguish past-week craving from episode-level urge. [29]
Perceived attention control	Attentional Control Scale, original 20 items	Focusing and shifting; later validation N=728. Self-report is not behavioral attention performance. [83, 84, 85]
Positive digital wellbeing	Digital Wellbeing Questionnaire, 13 items	2026 English validation N=1,853 US/UK young adults; agency, resilience, social connection and communion. Promising but longitudinal responsiveness remains uncertain. [86]

SMD and BSMAS cover control-failure or compulsive-use symptoms; neither is a standalone diagnostic test. [80, 81]

Do not use every questionnaire as a co-primary outcome. A practical research battery would select one problem-severity instrument appropriate to the population, one mechanism measure, and brief functional/utility outcomes; positive wellbeing can be secondary. Preserve published wording and recall periods, or label and validate modifications. Internal consistency alone cannot establish diagnosis, responsiveness or cultural equivalence. [79, 80, 14, 86, 17]

Self-reported screen time and logged behavior

Parry and colleagues found only moderate correspondence between reported use quantity and logs, $r=.38$, and weaker correspondence between problematic-use scales and logged quantity, $r=.25$. The first analysis included 66 effect sizes from 44 studies; these are not 66 independent trials. Self-report cannot simply replace logging of behavior. [87]

Conversely, low duration-symptom correlations do not make a symptom scale invalid merely because impairment is not time. Ellis and colleagues found generally weak relationships between use scales and phone logs, while an attention-control validation found little convergence with antisaccade performance. Reports of distress, agency and intention answer different questions from device records. [88, 85]

Behavioral telemetry dictionary for the proposed studies

Metric	Operational definition to preregister	Interpretation or coverage problem
App opens and transitions	Foreground entry after a fixed inactivity/background threshold; distinguish attempted from completed opens	A blocked attempt and a successfully opened app are different events; external links can trigger entries.
Unlocks and pickups	Define separately using available device events or system summaries	These are not interchangeable; a check can occur without unlocking or opening the target app.
Sessions and duration	Foreground active time with a declared idle timeout; sensitivity to threshold	Long useful reading and idle foreground time can inflate totals.
Total and social-media time	Predefined app/site category and measured devices, including the intermediary	Category migration and desktop/browser substitution can hide displacement.
Task success and completion time	Accuracy, completeness and timeliness against the stated need; explicit completion marker	User satisfaction alone cannot establish factual retrieval success.
Post-completion continuation	Active minutes or interactions after task completion	Objective proxy; chosen leisure after completion is not automatically unwanted.
Feed interactions and scrolling	Impressions, swipes, distance and time under an explicit viewport rule	Pixels vary by device and layout; scrolling is not proof of attention or comprehension.
Notification interactions	Delivered, surfaced, dismissed and opened alerts, where observable	Delivery does not guarantee attention; previews may satisfy a need or induce another check.
Rechecking and short-return loops	Return to the same source/need within a prespecified window	May represent uncertainty, conversational turn-taking or habit; requires context.
Nighttime use	Use during a participant-defined sleep opportunity	Shift work, caregiving and chosen schedules make fixed clock windows misleading.

This dictionary is a proposed instrumentation plan, not a claim that all operating systems expose these events. Test availability and reliability on each supported device. Missing logs must remain missing, not become zero use. Pair telemetry with sparse, near-time intention/valuation reports, recognizing that prompting itself can change behavior. The distinction between objective time and subjectively classified unwantedness is essential. [87, 35, 32]

9 Moderators and vulnerable populations

Population or characteristic	What is supported	Product implication and uncertainty
Adolescents	Habit/task-delay associations vary within and between individuals; problematic and intense use differ across countries. [34, 13]	Do not extrapolate an adult prototype trial to adolescents. Include social belonging and developmental context.
University students	Many interface and reduction studies recruit students. [50, 51, 1]	Convenient feasibility population; weak basis for general-population effect sizes.
Adults	Large field experiments and broad age-range experience sampling improve scope, but remain selected by platform and consent. [3, 35, 38]	Recruit beyond motivated early adopters; measure work, care and community use.
ADHD	Longitudinal review suggests potentially reciprocal associations, more consistent for problematic use; recent cohort associations remain observational. [89, 90]	No demonstrated differential benefit from extraction. Attention scaffolding and burden should be co-designed and tested.
Anxiety and depression	Problem-use symptom associations are repeatedly reported, including meta-analytic evidence. [81, 91]	Confounding, common-method overlap and reverse causation remain. Treatment-effect moderation requires randomized interaction tests.
High FOMO	Associated with engagement; notification suppression can increase situational FOMO, while trait FOMO did not moderate behavior in one RCT. [25, 42]	Test predictable delivery and trusted coverage; do not assume high-FOMO users benefit most.
Heavy users	Intensity is not interchangeable with impaired control; restrictions may remove valued activity. [13, 3]	Recruit on repeated unwanted use and goal conflict, not hours alone.
Culture and geography	A 29-country adolescent study found contextual variation in intensity associations. [13]	Validate translated measures and delivery expectations; US/UK or European results are not universal.

The ADHD review included 28 longitudinal studies, but no clinically referred diagnosed-ADHD samples; only a minority of studies testing both directions supported reciprocal associations. A 2026 ABCD analysis followed 8,324 children and associated social-media use with later inattention, with no detected moderation by ADHD diagnosis, medication or genetic risk. It did not test differential

intervention benefit and does not show that phones cause ADHD. [89, 90]

Subgroup claims should distinguish three propositions: a group has higher baseline difficulties; an exposure association differs by group; an intervention works differently by group. Evidence for the first does not establish the third. The proposed program should preregister a small set of interaction tests and report uncertainty, rather than interpret one subgroup's significant result and another's nonsignificant result as a difference.

10 Substitution effects

Substitution is an empirical question, not an inevitability or a negligible nuisance. Digital Addiction found material displacement from limited target apps into other apps. By contrast, HabitLab's randomized variation in intervention frequency did not detect a corresponding increase in other measured apps/sites or linked devices, although only 82 users contributed linked cross-device data. [3, 74]

These findings are compatible: intervention content, motivation, available alternatives and measurement coverage differed. One trial cannot establish that Instagram-to-YouTube, TikTok-to-browser, social-to-news or phone-to-computer migration never occurs. The evidence is **Mixed**, with credible examples of both displacement and broader reductions. [3, 74]

The intermediary itself belongs in the accounting boundary. Count its opens, repetitive queries, digests, reading time and transitions back to sources. Lower original-app minutes with unchanged ecosystem unwanted use is a product failure under the central hypothesis. This is a proposed evaluation rule, not a finding already established for extraction systems.

Also distinguish beneficial reallocation from harmful displacement. Moving from aimless scrolling to a deliberately selected article can increase browser time without undermining wellbeing; a new compulsive news-checking loop can reduce social-app time without solving the problem. Retain intention, experienced value and missed obligations alongside device totals. [32, 11]

11 Evaluation of the product hypothesis

The hypothesis contains several distinct causal claims. The table gives review judgments for each; a positive result on an early step is not sufficient for the whole product.

Testable mechanism	Supporting evidence	Challenge or alternative	Grade and design implication
A specific information need can be represented and served elsewhere	Bonsai and TwitInfo demonstrate related capabilities. [5, 70]	Exploration, belonging and reciprocal interaction may not be reducible to a query. [77, 32]	Preliminary. Begin with explicit, verifiable tasks and measure coverage failures.
External delivery reduces source-app entry	Entry friction and selective interfaces change access behavior. [53, 1]	Users can still open the source for reassurance or another motive. No direct general extraction test located. [42, 9]	Preliminary indirect support. Measure transitions after delivery and failed retrievals.
Avoiding source cues reduces continuation	Feature and feed experiments change engagement. [4, 6]	Not every cue manipulation works; internal urges and existing routines persist. [40, 30]	Moderate for particular features; preliminary for this mediation chain. Randomize cue exposure separately.

Testable mechanism	Supporting evidence	Challenge or alternative	Grade and design implication
Reduced continuation is genuinely less unwanted use	Regret and goal-conflict studies provide relevant constructs. [32, 35]	Unplanned discovery or restorative leisure can be valued. [33]	Preliminary. Use near-time valuation rather than label all excess minutes harmful.
Utility is preserved better than under blocking	Selective-function and adaptable-interface prototypes are promising. [2, 8]	Information, social context and interaction may be lost; no qualifying direct superiority test located. [1, 72]	Insufficient. Require task and social-utility noninferiority, not merely satisfaction.
Reduced source use persists without substitution	Some field reductions extend beyond target apps. [74]	Other experiments show displacement and rebound. [3, 50]	Mixed. Include intermediary, browser and other devices, plus follow-up.
Behavioral changes improve wellbeing or functioning	Some restriction RCTs show benefits. [49, 50]	Detox synthesis and intervention trials contain null or dissociated outcomes. [47, 64]	Mixed. Measure outcomes directly and keep them separate from time.

Inference: the strongest causal proposition is that digital environments can influence behavior. The weakest product-specific proposition is that automatically serving requested information preserves the full range of benefits while durably removing unwanted engagement. Existing evidence warrants a staged research program, not a claim of demonstrated efficacy.

12 Potential failure modes and safeguards

The risks below are hypotheses to monitor. Evidence supports some underlying mechanisms, while the exact safeguards are mostly design proposals requiring evaluation.

Failure mode	Empirical basis or uncertainty	Proposed safeguard and test
A second compulsive feed	Fast informational rewards can reinforce checking. [15]	Finite results, no unsolicited continuation, explicit completion. Randomize result bounds and measure intermediary checking.
More notification checking or FOMO	Blanket suppression can increase FOMO without reducing logged checks. [42]	Predictable user-set schedules and clearly defined urgent exceptions; test missed-information rates and unscheduled checking.
More total information consumption	Lower retrieval cost could expand demand; direct rebound evidence for extraction is absent.	Cap recurring queries by user choice; show source scope and measure total ecosystem use. Do not assume faster retrieval reduces demand.
A hidden replacement recommender	Intentional feed building can still produce an ongoing feed. [5]	Make source selection and ranking inspectable; optimize task completion, not time or click-through. Test whether users understand selection rules.
Habit moves to a new interface	Device/context habits are not limited to one feed. [30]	Count repetitive low-value queries and returns, not only original-app opens. Evaluate delayed availability or a digest when appropriate.
Meaningful interaction is lost	Regret and value vary within an app; removing features can omit valued information. [32, 1]	Preserve replies, context and access to chosen people; offer transparent source opening when needed. Measure connection and relationship consequences.
Important information is filtered out	Reader-mode extraction can omit essential content. [72]	Evaluate recall of priority items, timeliness and unsupported sources, with a visible completeness boundary.

Failure mode	Empirical basis or uncertainty	Proposed safeguard and test
AI summaries distort meaning or encourage overreliance	Factuality and social-framing experiments establish relevant risks. [76, 75]	Keep attribution, dates, links and optional original text; audit omissions and contradictions. Do not infer fidelity from fluent prose or user confidence.
Friction becomes irritating or inaccessible	Interface friction can worsen experience and increase abandonment. [45, 40, 73]	Adjustable controls and explicit overrides; include cognitive/access burden, disablement and dropout as outcomes.
Shame or addiction identity reduces agency	Addiction framing reduced perceived control in an experiment. [92]	Describe specific unwanted behaviors and participant-chosen goals; avoid diagnosing users from app data. Test agency and self-blame.

Hiding comments, likes or follower counts by default should therefore be an adjustable experiment rather than a universal prescription. Removing a feature can remove a trigger, useful context, or social participation at the same time; the relevant net outcome has not been demonstrated for each component. [40, 32]

13 Product principles and their evidential status

“Evidence-backed” here means supported for a related measured behavior or experience, not that the exact product is validated.

Principle	Status	Concrete implementation implication
Preserve user-selected functions while changing exposure	Supported design direction; preliminary efficacy [2, 1]	Define supported tasks and retain necessary actions; compare with app-wide restriction.
Make continuation an explicit choice	Moderate proximal evidence [53, 4]	No automatic next item; an obvious finish action and optional continuation.
Let users choose intent and mode	Preliminary agency evidence [8, 78]	Support specific retrieval and explicitly chosen exploration; do not classify all exploration as failure.
Provide finite result sets and stopping points	Plausible, partially supported by feature experiments [40]	Show a completed set with scope/date; test finite versus continuous availability separately.
Use scheduled or batched delivery when suitable	Mixed-to-moderate evidence [55, 58, 42]	User-defined cadence, urgent exceptions, no assumption that three batches/day is universally optimal.
Offer light, purposeful friction	Moderate for immediate access decisions; durability uncertain [53, 54]	Delay or confirm repeated nonessential entry; measure annoyance, overrides and task cost.
Prefer explicit sources and transparent ranking	Preliminary prototype support [5]	Show which sources were searched and why an item appears; evaluate effort and trust.
Offer chronology as a choice	Strong evidence of engagement change in one large platform experiment; mixed generalization [6, 38, 39]	Do not advertise chronology as inherently healthy; separate ranking from unlimited supply.
Remove engagement counts or recommendations selectively	Component-specific evidence remains limited [40, 78]	Treat likes, followers, comments and suggested items as separate randomized factors.

Principle	Status	Concrete implementation implication
Use text-first summaries only when they serve the task	Reasonable but untested wellbeing hypothesis [72, 75, 76]	Preserve media or originals when meaning depends on them; test accuracy, utility and workload.

Avoid an optimization target that rewards more queries, more opened digests or more time in the intermediary. The suggested target is successful useful access with a low burden of unwanted engagement, subject to fidelity and social-utility constraints. This is a proposed objective, not a validated universal wellbeing score.

14 Proposed R and D experimental program

All durations, thresholds and sample sizes below are planning proposals. They are not known optimal doses or empirical extraction-product effect sizes. Preregister the product version, eligibility, estimands, primary outcomes, exclusions, missing-data rules and analysis before examining treatment outcomes.

Primary success criteria and central metric

Use two co-primary endpoints: **utility preservation** and **unwanted engagement**. Require utility to be noninferior to usual access and unwanted use to improve. The phrase “preserved utility” requires a justified acceptable-loss margin; a nonsignificant superiority test cannot demonstrate it. [12]

Define an eligible information-need episode independently of successful product use. In a task substudy, use assigned requests; in the field, combine prospectively saved intentions with probability-sampled reports of natural needs. Include blocked, abandoned, late and unsupported needs. Otherwise the product could improve its score by discouraging difficult requests or deleting useful activity from the denominator.

For each person-week, define:

- **K**: total prespecified importance weight of eligible needs. Weights should be bounded and set before the result is seen.
- **D**: the sum of those weights multiplied by validated success scores between zero and one, including accuracy, completeness and timeliness; $0 \leq D \leq K$.
- **U**: unwanted active-use minutes across the source services, intermediary and measurable substitutes. Logs provide duration; near-time reports and prior intentions classify unwantedness. This is a **hybrid endpoint**, not fully objective telemetry.
- **tau > 0**: a prespecified time scale, for example one minute per utility-weight unit, used for an exploratory summary score.

Report **utility retention** $R = D/K$ and **unwanted burden** **U** separately. A bounded exploratory efficiency score is:

$$Q = D / (K + U/\text{tau}), \text{ for } K > 0.$$

Q approaches one only when desired utility is obtained with little unwanted engagement. It remains finite when U is zero, gives zero when no desired utility is delivered, and penalizes failures through K. It is unvalidated, depends on the chosen scale tau, and requires sensitivity analyses. K=0 weeks should be reported separately; keep their unwanted minutes in engagement analyses. A raw D/U ratio

is undefined at zero unwanted time and should not be primary.

Also report absolute D, K and U, unsuccessful or suppressed needs, important-message omissions, chosen recreation, social connection and satisfaction. More content is not automatically more utility. Weighting schemes must not permit many trivial successes to conceal one unacceptable high-priority miss. Analyze people first, rather than letting the most active query users dominate a pooled event ratio.

Feasibility stage before the efficacy experiments

Run a two-to-three-week feasibility study with approximately 20-30 diverse users and a benchmark covering specific accounts, messages, communities, news, events, videos and questions. Include tasks requiring a reply, images, conversational context and time-sensitive information. Masked assessors should check a consented sample for factual accuracy, omission and latency; interviews should identify needs participants could not formulate or were discouraged from attempting.

Proceed only for task categories meeting a prespecified utility threshold. A system unable to retrieve an important message has an implementation/coverage failure; this does not by itself disprove the psychological cue-removal mechanism. Conversely, excellent benchmark retrieval does not establish reduced unwanted use.

Experiment A App entry prevention and cue exposure

Core comparison: ordinary source-app access versus external delivery of the same desired content. Use counterbalanced tasks in an initial laboratory study, then randomize participants in a field deployment. Match task difficulty, freshness, content and latency where possible. Give free-choice time after completion so continuation can be observed.

Mechanism condition: add a deep link to the requested item with surrounding feeds/recommendations hidden, where feasible. This separates avoiding app launch from avoiding surrounding stimuli. Randomizing cue availability after delivery gives stronger mechanism evidence than comparing two interfaces that also differ in relevance, length and friction.

Measure source opens, actual cue impressions, retrieval success, completion latency, post-completion activity, rechecking, near-time unwantedness, craving and satisfaction. Identify the randomized total effect first. Fewer opens correlated with fewer unwanted minutes is not sufficient causal mediation: craving or task difficulty can influence both. Mediation analyses need additional assumptions and sensitivity tests. [93]

Experiment B Search versus feed

Use a **2 × 2 design**: query-driven retrieval versus feed presentation, crossed with finite stopping points versus continuous availability. Use the same source corpus, content types, goal and freshness; match relevance and ranking opportunities as far as possible. Record unavoidable differences explicitly.

Include specified information tasks and separately chosen open-ended recreation. Outcomes should include useful completion, effort, enjoyment, valued discovery, voluntary continuation and regret. A narrow query compared with an unrelated unlimited feed would confound interaction mode with supply and relevance.

A simple-effect question is whether querying adds value within finite presentation; the interaction asks whether its effect differs between finite and continuous conditions. If finite results account for

the benefit, simplify the product. If the query interface improves agency while increasing useful time, that may still be a success under the proposed metric. Short crossover studies can identify immediate mechanisms; use a parallel field design to assess durable habits and avoid carryover.

Experiment C Extraction versus blocking

Randomize users to three arms after two baseline weeks. A proposed main schedule is eight intervention weeks and four follow-up weeks, with the final intervention weeks as the primary assessment window.

Arm	Configuration	Main interpretive purpose
Usual access	Existing access plus the same low-burden measurement	Assignment-level effect relative to observed ordinary use
App blocker	Credible user-configured limits or schedules, with visible override	Benchmark against an existing restriction strategy
Extraction	Desired content delivered externally; original access remains possible	Added value and real-world adherence of the intermediary

Equalize onboarding and measurement. If extraction also blocks or redirects source entry, call it extraction-plus-redirection and add a component comparison when feasible. Do not attribute its entire effect to information delivery.

The first claim requires unwanted-use superiority and utility noninferiority versus usual access. The stronger differentiation claim requires utility superiority over blocking and unwanted-use noninferiority to blocking. The blocker must itself work in this sample for that latter comparison to be informative. Nonadherence can artificially favor apparent noninferiority; show assignment-based estimates and carefully specified adherence sensitivity analyses. [12]

A 60-90-person three-arm pilot is reasonable for feasibility, data completeness, variance and adherence estimation. It should not be treated as a definitive efficacy trial.

Experiment D Delivery schedule and recurrent checking

Compare user-selected digest delivery, on-demand access and a carefully bounded adaptive prompt strategy. A micro-randomized trial can randomize support at prespecified eligible moments to estimate proximal effects and how they vary with context. Define availability, prompt ceilings, outcome windows and repeated-observation analysis in advance. The method is established; its efficacy for this product is not. [67, 68]

Include missed urgent information, unscheduled checks, FOMO and burden. A digest that reduces source entry but increases intermediary monitoring fails the intended mechanism. Test a simple schedule before assuming machine-learning adaptation is necessary.

Experiment E Durability substitution and social utility

Extend follow-up to evaluate novelty loss, voluntary abandonment and rebound. Observe target apps, browser versions, competing apps and the intermediary; add consented desktop logging where possible. Use interviews to explain displacement and valued uses lost. If coverage is incomplete, restrict claims to measured devices.

Sample relationship tasks and conversations, not only fact lookup. Compare whether users remain appropriately informed and able to participate. Assess the effect of summary errors on subsequent actions rather than asking only whether the summary looked useful.

Power and analysis

Power the confirmatory trial for both co-primary requirements, using pilot estimates of person-level variance, baseline correlation, episode clustering, skew, zero-use days, attrition and uptake. Simulate joint power. Hundreds of episodes from a few people are not hundreds of independent participants.

For illustration, a two-group normal approximation with two-sided alpha .05, 80% power and $d=.30$ gives approximately **175 analyzable participants per arm**, or about **219 recruited** allowing 20% loss. If two pairwise contrasts each use two-sided alpha .025, the corresponding approximation is **212 analyzable**, about **265 recruited per arm**; a three-arm study would recruit about 795. These are planning calculations, not predictions from extraction trials.

Utility can require more participants. At one-sided alpha .025 and 80% power, a continuous utility score with SD .15, true difference zero and a .05 noninferiority margin requires about 142 analyzable participants per arm under the simple independent-means approximation. A single binary success outcome near .85 with a five-percentage-point margin requires about 801 per arm. Repeated tasks require a clustered calculation. The acceptable margin must come from user needs and consequences, not affordability.

Use intention-to-treat analysis, baseline adjustment and prespecified randomization strata. Report interpretable absolute effects and confidence intervals; use models appropriate to skewed counts/minutes and within-person clustering. Missing records are not zero use, and uninstalling is an outcome rather than automatic exclusion. Report missingness by arm and conduct sensitivity analyses for outcomes among dropouts. Separate confirmatory endpoints from exploratory mechanisms and moderators; control multiplicity through a prespecified hierarchy or appropriate adjustment.

No participant or researcher can realistically be blind to all interface differences. Use masked utility adjudication and objective logging where feasible, keep contact comparable, disclose developer conflicts, freeze product versions and publish an analysis plan. Include qualitative interviews with enthusiastic users, nonadopters, dropouts and participants who lost utility.

Falsification and decisions

- **Mechanism weakened:** cue exposure falls, yet unwanted use does not, with confidence intervals excluding a worthwhile benefit.
- **Utility claim fails:** accurate/timely access, valued interaction or chosen recreation crosses a prespecified unacceptable-loss threshold despite lower time.
- **Displacement failure:** source-app reductions are offset by unwanted intermediary, browser, other-app or desktop use.
- **Durability failure:** improvement disappears late in treatment or rebound erases the gain.
- **Differentiation failure:** extraction adds no utility over a functioning blocker, or simple bounded/deep-link access provides equivalent benefit with lower burden.
- **Inconclusive:** confidence intervals allow meaningful benefit and important harm. This is not evidence of equivalence or a decisive falsification.

These rules distinguish failure of one implementation from failure of the theory. A low-fidelity retriever cannot fairly test utility-preserving cue avoidance; a high-fidelity system that reduces cues without reducing unwanted use is a much stronger challenge to the central mechanism.

Publication status corrections and uncertainty

Guess et al. has Instagram-data errata dated 5 December 2024 and 19 March 2026, not a retraction. The authors state key treatment estimates should be unaffected. This review avoids precise corrected exposure claims. [6]

Olson et al. has a published correction linked at DOI 10.1007/s11469-022-00898-8; consult it with the original. Sohn et al. has author-name corrections, reflected in this bibliography. [52, 91]

Bonsai, the Netflix autoplay paper, Time2Stop and MindShift have peer-reviewed final publications despite accessible arXiv or author-manuscript versions. Such access copies are not counted as independent studies. [5, 4, 62, 63]

No preprint-only result is used as a principal efficacy foundation. Retraction status has not been certified through an exhaustive registry audit. Ambiguous sample details remain labeled NR, and critical narrative/framework papers are identified as theory or review rather than experiments. The bibliography is a traceable record of the sources used, not a guarantee that every relevant paper was retrieved.

Evidence table 1. Problematic use, habits and individual differences

Study	Year	Design	N	Population	Intervention / exposure	Objective outcome?	Main finding	Limitations	DOI / link
Oulasvirta et al. [15]	2012	Longitudinal logging; small intervention	136 phone; 160 laptop; intervention 15; diary 12	US/Finnish device users	Repeated checking; dynamic address-book content	Yes: use logs; diaries	Brief checks opened pathways to other applications; dynamic content increased checking.	Early devices; small intervention; not modern feeds.	DOI
Panova & Carbonell [16]	2018	Critical narrative review	N/A	Smartphone-use literature	Addiction versus problematic-use framing	No new data	Evidence insufficient to establish distinct smartphone addiction.	Narrative review; does not imply impairment is absent.	DOI
de Segovia Vicente et al. [35]	2024	Preregistered 2-week ESM and logging	1,315; 691 with logs	Flemish adults, 18-82	Goal-less social-app use	Partial: duration logged; mindlessness reported	Scrolling correlated with goal conflict and guilt; wellbeing association very small.	Observational; mindlessness not objectively detected.	DOI
Meier et al. [34]	2023	3-week experience sampling	312; 22,809 assessments	Dutch adolescents, 13-15	Automaticity, checking and task delay	No: self-report	Habitual use associated with task delay, with substantial individual variation.	Noncausal; shared self-report measurement.	DOI
Meier [31]	2022	5-day diary	532; 2,331 diaries	German students	Checking habits, urges and interruptions	No: self-report	Checking linked to procrastination; direct wellbeing associations small or absent.	Reverse causation and common-method bias possible.	DOI
Anderson & Wood [92]	2025	Preregistered survey and experiment	380 survey; 824 experiment	US Instagram users	Addiction labels and randomized framing	No: perceived control/self-blame	Addiction framing reduced perceived control and increased self-blame.	Proximal reports; screening does not establish diagnosis.	DOI
Lemahieu et al. [94]	2026	Cross-sectional survey; SEM	944	Belgian adults	Habit and disconnection difficulty	No: self-report	Habit related to disconnection difficulty, but not disconnection behavior.	Cross-sectional; reported behavior and experience differ.	DOI
Ross et al. [30]	2025	2-week sensing and trip surveys	419	Columbus, Ohio Android users	Spatial familiarity and phone habits	Yes: GPS/app logs	Phone-use habits appeared stronger in familiar places.	Single city; observational; repetition is not automaticity.	DOI
Brand et al. [20]	2019	Integrative theoretical review	N/A	Addictive-behavior literature	I-PACE: cues, affect, control and learning	No new data	Proposes interacting vulnerabilities and learning processes sustaining problematic behavior.	Model, not a universally validated causal sequence.	DOI
Kardefelt-Winther et al. [22]	2017	Conceptual definition paper	N/A	Behavioral-addiction literature	Impairment-based diagnostic boundaries	No new data	Prioritizes sustained impairment and alternative explanations over symptom counting.	Proposed framework; not diagnostic consensus.	DOI

Study	Year	Design	N	Population	Intervention / exposure	Objective outcome?	Main finding	Limitations	DOI / link
Billieux et al. [21]	2015	Narrative review; pathway model	N/A	Problematic phone-use literature	Reassurance, impulsivity and extraversion pathways	No new data	Similar excessive use may arise through different mechanisms.	Hypothesized pathways; not validated diagnostic subtypes.	DOI
Sohn et al. [91]	2019	Systematic review/meta-analysis	41 studies; 41,871	Children/young people; study mean ≤25	Questionnaire-defined problematic smartphone use	Mostly no: reports	Problematic-use scores associated with depression, anxiety and sleep problems.	Mostly cross-sectional; heterogeneous cutoffs; no causal or diagnostic prevalence inference.	DOI
Boer et al. [13]	2020	Cross-national cross-sectional survey	154,981; 29 countries	Adolescents; mean 13.5	Intense versus problematic social-media use	No: self-report	Problematic use consistently accompanied poorer wellbeing; intense-use associations varied by country.	Cross-sectional; no universal time threshold established.	DOI
Thorell et al. [89]	2024	Longitudinal systematic review	28 studies	Children/adolescents, 0-17	Digital-media use and ADHD symptoms	Mostly reports	Associations ran both directions; more consistent for problematic use than duration.	Observational; common-rater bias; no clinically referred ADHD samples.	DOI
Nivins et al. [90]	2026	4-year ABCD cohort	8,324	US children; baseline mean 9.9	Social media, gaming and TV/video	No: use/symptom reports	Social-media use predicted increasing inattention; gaming and TV/video did not.	Observational; symptom change is not new ADHD diagnosis.	DOI

Evidence table 2. Engagement mechanisms and platform design

Study	Year	Design	N	Population	Intervention / exposure	Objective outcome?	Main finding	Limitations	DOI / link
Lindström et al. [24]	2021	Trace modeling and online experiment	>4,000 trace users; 176 experiment	Social-platform users/online participants	Like rewards; experimentally varied reward rate	Yes: posting behavior	Reward rate causally influenced posting timing in the experiment.	Production rather than passive scrolling; no dopamine measurement or addiction diagnosis.	DOI
Tam & Inzlicht [10]	2024	7 experiments; 6 preregistered	1,223 total	Student and nonstudent samples	Switching versus continuous digital viewing	Partial: switching behavior; boredom reported	Boredom encouraged switching; switching often increased boredom and lowered engagement.	Generalization weaker for articles/nonstudent samples; short tasks.	DOI
Bartoli & Benedetto [43]	2022	Between-group interface experiment	1,095 recruited; 1,009 analyzed	Italian smartphone users	Badge placement on static home screens	Yes: one first click	Badged icons attracted more first clicks.	Static, brief task; did not isolate badge color or test chronic checking.	DOI
Schaffner et al. [4]	2025	Randomized field experiment	76; 38/arm	US Netflix users	Autoplay off versus on, 10-17 days	Yes: donated watch history	Disabling autoplay reduced relative daily viewing and session duration.	Small, short study; baseline modeling unusual; Netflix does not establish short-video effects.	DOI
Alin et al. [40]	2026	Within-person laboratory experiment	48	TikTok-prototype participants	Friction for autoplay, scrolling, social investments	Yes: interactions and eye tracking	Autoplay/scroll friction changed engagement proxies but worsened experience; agency benefits inconsistent.	Short prototype; proxies do not establish clinical improvement.	DOI
Sherman et al. [44]	2016	Controlled scanner experiment	34 enrolled; 32 fMRI	Adolescents, 13-18	Experimentally assigned Instagram like counts	Yes: choices and fMRI BOLD	Popular images attracted likes; own highly liked photos increased reward-region response.	Small simulation; BOLD is not dopamine release or addiction evidence.	DOI
Chiossi et al. [36]	2023	Controlled pre/post laboratory experiment	60; 15/condition	Adults, 19-34	10-minute TikTok/Twitter/YouTube/rest break	Yes: prospective-memory task	TikTok condition showed worse post-break prospective memory.	Preference-linked assignment; platform/content confounds; no evidence of permanent impairment.	DOI
Leng et al. [27]	2019	Extreme-group cue-reactivity experiment	60 selected from 169	Chinese university participants	Social-network image/word versus neutral cues	No: craving/arousal ratings	Social cues increased reported craving chiefly among high-score participants.	Nonclinical grouping; fixed neutral-first order; no objective use outcome.	DOI

Study	Year	Design	N	Population	Intervention / exposure	Objective outcome?	Main finding	Limitations	DOI / link
Guess et al. [6]	2023	3-month platform field RCT	23,391 Facebook; 21,373 Instagram*	US adult platform users	Chronological versus algorithmic feeds	Yes: platform logs; attitudes reported	Chronological feeds reduced time/activity; political attitudes and knowledge did not detectably change.	Election context; no compulsivity measure; Instagram-data errata.	DOI
Gauthier et al. [38]	2026	Preregistered 7-week field RCT	6,043 baseline; 4,965 post	US adult X users	Algorithmic versus chronological feed	Partial: content/follows logged; engagement reported	Algorithm-on increased reported engagement; algorithm-off effect weaker; wellbeing unchanged.	Imperfect compliance; asymmetric effects; engagement was not screen-time telemetry.	DOI
Brady et al. [39]	2026	Registered 8-week feed RCT	2,000 randomized	Bluesky study participants	Engagement, chronological or diversified ranking	Yes: content/engagement logs	Ranking changed content quality and norms without significant own-engagement changes.	Election/custom-platform setting; no query-only or cross-device endpoint.	DOI

Facebook and Instagram counts are separate platform samples and must not be summed as unique people.

Evidence table 3. Digital-wellbeing interventions

Study	Year	Design	N	Population	Intervention / exposure	Objective outcome?	Main finding	Limitations	DOI / link
Monge Roffarello & De Russis [7]	2023	Systematic review/meta-analysis	62 articles; pooled 255	Digital self-control tool users	Timers, blockers, monitoring and design changes	Mixed: time measures	Pooled time reduction $g=0.47$.	Mostly short studies; synthesis uses within-treatment changes, not pooled randomized contrasts.	DOI
Radtke et al. [46]	2022	Systematic review; 12 RCTs	21 studies; 3,625	Mostly adult digital-media users	Temporary full/selective abstinence	Mixed behavior and reports	Wellbeing, relationship and self-control findings mixed, including null and negative outcomes.	Heterogeneous treatments, outcomes and adherence; no pooled causal estimate.	DOI
Lemahieu et al. [47]	2025	Preregistered systematic meta-analysis	10 studies; 4,674	Mostly Western adult/student samples	Temporary social-media abstinence	Mainly no; one objective-use study	No significant pooled benefit for affect or life satisfaction.	Few studies; inconsistent adherence; average null does not prove equivalence.	DOI
Zhang et al. [48]	2026	Umbrella review/meta-meta-analysis	29 meta-analyses; 66,530 reported*	Internet/phone/gaming problem-use groups	CBT/other psychological, exercise, multimodal treatment	Primarily symptom scales	Smartphone symptom effect SMD -0.80 ; certainty generally low.	High heterogeneity, overlapping reviews and mixed constructs; not consumer-tool efficacy.	DOI
Allcott et al. [3]	2022	Preregistered factorial field RCT	2,126 randomized	US adult Android volunteers	Incentives and optional delayed-change app limits	Yes: app logs; other devices reported	Limits reduced target use 22 min/day, with 12 minutes shifting elsewhere.	Opt-in pandemic sample; economic model differs from clinical diagnosis.	DOI
Castelo et al. [49]	2025	Preregistered delayed-intervention RCT	467 randomized; 313 complete; 119 compliant	Adult iPhone users	2-week mobile-internet block; calls/texts retained	Yes: compliance, screen time, attention task	Blocking improved sustained attention and reported mental health/wellbeing.	Low adherence; complete-case primary analyses; no active control.	DOI
Pieh et al. [50]	2025	Parallel RCT; follow-up	111; 58/53	Healthy students; mean 22.7	≤ 2 hours/day smartphone use for 3 weeks	Device screenshots; symptoms reported	Short-term mood, stress, sleep and wellbeing benefits; use rebounded.	Small unblinded student trial; follow-up attrition; utility unmeasured.	DOI
Olson et al. [52]	2023	Feasibility study and active-control RCT	51 feasibility; 82 recruited/70 analyzed RCT	Young adult iPhone users	Ten settings/environment strategies	Device screenshots and cognition tasks	Bundle reduced time/problem scores and improved sleep; depression contrast null.	Components inseparable; adherence exclusions and baseline imbalance.	DOI

Study	Year	Design	N	Population	Intervention / exposure	Objective outcome?	Main finding	Limitations	DOI / link
Zimmermann & Sobolev [51]	2023	Preregistered 3-arm field RCT	112	Postgraduate iPhone users	Grayscale versus limits versus self-monitoring	Yes: device-recorded duration	Grayscale reduced time immediately; limits gradually; wellbeing/academic outcomes null.	Short, selected sample; reduced time does not establish preserved utility.	DOI
Keller et al. [60]	2021	20-day active-control RCT	232 enrolled; 110 post; 88 follow-up	Adult iPhone volunteers	Goal/values training versus daily detox plan	Device values manually transcribed	Both arms improved; goal training not superior; unlocks unchanged.	High attrition; no untreated group; manual data transfer.	DOI
Brailovskaia et al. [95]	2023	3-group controlled intervention	619; 200/226/193	German smartphone users	7-day abstinence versus 1-hour/day reduction	No: online surveys	Both improved reported outcomes; reduction often yielded more persistent benefits.	Randomization details unverified here; self-report; optimal dose unestablished.	DOI
Orzikulova et al. [62]	2024	8-week controlled field experiment	71	Smartphone users	Time2Stop adaptive timing and explanations	Yes: app-visit logs	Adaptive variants reduced visits about 7-8.9% and improved receptivity.	Modest prototype sample; component comparisons; no clinical or extraction test.	DOI
Wu et al. [63]	2024	5-week comparative field experiment	25 field; formative 12 and 10	Smartphone users	MindShift context-sensitive LLM persuasion	Yes: usage telemetry	Tailored variants reported 7.4-9.8% lower duration and better acceptance.	Small short prototype; persuasion is not retrieval; acceptance is not utility.	DOI
Throuvala et al. [65]	2020	10-day online RCT	143	UK university students	Mindfulness, monitoring and mood-tracking apps	No: main outcomes reported	Distraction improved; habitual behavior, nomophobia and social-media time did not.	Short bundled intervention; mindfulness component not isolated.	DOI
Mertens et al. [64]	2026	Preregistered 3-week RCT	70; 35/arm	Adult iPhone users	Wellspent personalized full-screen limit reminders	Device minutes manually entered	Target-app time fell about 29 min/day; problematic social-media score unchanged.	Small, short trial; manual transfer; developer conflicts disclosed.	DOI
Allcott et al. [69]	2020	Randomized 4-week field experiment	2,743 price-offer; 1,661 impact sample	US adult Facebook users	Paid deactivation versus control before 2018 election	Profile compliance/news quiz; wellbeing/time reported	Deactivation improved subjective wellbeing but reduced factual news knowledge.	Selected willingness-to-accept subgroup; election-period context; most outcomes self-reported.	DOI

Umbrella-review participant totals may include overlapping primary studies. Device screenshots and manually transferred device values are distinguished from automatically collected telemetry. N/A = not applicable; EMA/ESM = momentary assessment; RCT = randomized controlled trial.

Evidence table 4. Notification and checking interventions

Study	Year	Design	N	Population	Intervention / exposure	Objective outcome?	Main finding	Limitations	DOI / link
Fitz et al. [55]	2019	3-week exploratory field RCT	237	Indian Android owners	Usual, hourly, 3/day or no notifications	Yes: unlock logs; reports	Three daily batches improved several reports; complete silence increased anxiety/FoMO.	Multiple outcomes; total-use data unreliable; hourly effects mostly null.	DOI
Kushlev et al. [56]	2016	Randomized 2-week crossover	221 completed	Canadian undergraduates	Alerts/phone nearby versus silent/phone away	No: symptoms reported	Interruption condition increased reported inattention/hyperactivity; direct wellbeing/productivity effects null.	Bundled alert/location treatment; questionnaire symptoms are not ADHD diagnosis.	DOI
Pielot & Rello [57]	2017	Counterbalanced 24-hour field experiment	30	White-collar workers	All-device alerts off versus usual delivery	No: experience reports	Less distraction and greater productivity accompanied worries and lower connectedness.	One day; selected sample; follow-up maintenance self-reported.	DOI
Stothart et al. [28]	2015	Randomized laboratory experiment	212 recruited; 166 analyzed	US undergraduates	Call/text notifications versus none	Yes: sustained-attention errors	Notifications increased task errors without phone interaction.	Brief laboratory task; exclusions; no evidence of lasting attention damage.	DOI
Dekker et al. [42]	2025	Preregistered randomized field trial	274 randomized; 205 analyzed	Android users, 18-30	Disable app alerts for 1 week	Yes: checking/screen-time logs	Checking and screen time did not significantly decline; situational FoMO increased.	Post-randomization compliance exclusions; short treatment; limited baseline problems.	DOI
Liao & Sundar [59]	2022	Cross-sectional observational study	138	iPhone users	Self-selected sound, vibration or silent modes	Yes: Screen Time/pickups	Silent-mode users checked more, especially with higher FoMO/belonging needs.	Nonrandomized; frequent checkers may select silent mode.	DOI
Kushlev & Dunn [58]	2015	Randomized 2-week crossover	124	Students and employed adults	Email ≤ 3 checks/day versus unrestricted	No: checking/stress reported	Restricted email checking reduced reported stress.	Bundled short regimen; not isolated alerts, clinical treatment or universal optimum.	DOI

Evidence table 5. Friction, limits and interface changes

Study	Year	Design	N	Population	Intervention / exposure	Objective outcome?	Main finding	Limitations	DOI / link
Grüning et al. [53]	2023	6-week field study and component RCT	280 field; 500 online	one sec users/online participants	Entry delay, dismissal choice and deliberation	Yes: opens; time reported	Field completed opens fell 57%; online delay/dismissal helped, deliberation text did not.	Field uncontrolled; developer involvement; total-use effects unclear.	DOI
Haliburton et al. [54]	2024	Longitudinal observational deployment	1,039; 249 survey	one sec users	App-entry intervention over mean 13.4 weeks	Yes: attempt/continuation logs	Attempts declined and users took periodic breaks.	Self-selection; no randomized control; attempts are not total screen time.	DOI
Ruiz et al. [45]	2024	Controlled interface experiment	30	Students, 21-31	Infinite feed versus required reaction/skip	Yes: recognition/time	Required interactions improved memory but lengthened browsing and increased frustration.	Small fictitious-feed task; exposure-time confound; no impairment outcome.	DOI
Meinhardt et al. [41]	2025	7-day deployment/context analysis	72 of 160 installers; 946 observations	US Android users reporting regret	Break overlay after 15-minute scrolling	Yes: scrolling and cessation latency	Context predicted stopping and prompt response.	No control; installer attrition; context not randomized.	DOI
Orzikulova et al. [2]	2023	16-day between-group field comparison	56 analyzed; 65 recruited	Mostly South Korean student Android users	FinerMe app- versus feature-level feedback/limits	Yes: usage logs	Use declined in both groups; feature targeting helped some passive features.	No untreated control; fixed phase order; utility preservation unproven.	DOI
Lyngs et al. [1]	2020	6-week 3-arm randomized field study	58	Student desktop Facebook users	Goal reminders or removed feed versus control	Yes: browser logs	Interventions reduced scrolling/use; some users reported FoMO or lost information.	Small exploratory trial; browser-only coverage; utility costs possible.	DOI
Lee et al. [66]	2025	2-week before/after deployment	29	Social/video website users	PurposeMode toggles for distracting patterns	Yes: browser logs; EMA	Distraction and logged time decreased.	Uncontrolled; bundled toggles; short deployment.	DOI
Hiniker et al. [61]	2016	Survey and 2-week field deployment	232 survey; 23 field	US Android users	MyTime user-defined low-value app limits	Yes: app logs	Low-value app time fell 21%; valued-app time unchanged.	Small uncontrolled study; unchanged time does not demonstrate preserved benefit.	DOI
Kovacs et al. [73]	2018	3 randomized browser experiments	217; 409 eligible/91 subset; 93	HabitLab browser users	Rotating/static interventions and explanations	Yes: use and uninstall logs	Rotation reduced time but increased attrition; explanations helped.	Selection/filtering; multiple interventions; retention is part of effectiveness.	DOI

Evidence table 6. Filtering, extraction and aggregation

Study	Year	Design	N	Population	Intervention / exposure	Objective outcome?	Main finding	Limitations	DOI / link
El Malki et al. [5]	2026	Exploratory 2-phase deployment	15	Mostly academic Bluesky users	Bonsai intent-authored custom feeds	Interaction traces; no total-use endpoint	Intent-based filtering feasible; authoring required effort and transparency.	Small, short, selected sample; still a feed; no blocker comparison.	DOI
Hong et al. [71]	2010	Prototype/format ive evaluation	15	Knowledge workers; 2 organizations	FeedWinnower faceted RSS/social aggregation	Usability only; no wellbeing endpoint	Demonstrated topic/person/source/time filtering across sources.	Small formative study; no causal use or displacement outcomes.	DOI
Marcus et al. [70]	2011	Algorithm and user evaluation	12 users; 1 journalist	Nonexperts and journalist	TwitInfo query-to-event summaries/timelines	Yes: event detection/task performance	Users produced meaningful event summaries rapidly.	No craving/time outcomes; real-time dashboard may invite monitoring.	DOI
Li et al. [72]	2019	Controlled reading experiment	391 readers; 1,100 pages characterized	Online readers; non-phone displays	Firefox Reader View versus normal pages	Yes: reading speed	Reader View increased reading speed 5%; extraction sometimes lost essential content.	No compulsivity outcome; mobile excluded; fidelity risk remains.	DOI
Govers et al. [75]	2026	4-condition online experiment	144	Adult online participants	AI narrative/percent age summaries of threads	Opinion responses; no usage telemetry	Percentages increased conformity; narratives produced false balance; engagement willingness unchanged.	Simulated threads; proximal responses; no long-term behavior evidence.	DOI

Evidence table 7. Search, feeds and agency

Study	Year	Design	N	Population	Intervention / exposure	Objective outcome?	Main finding	Limitations	DOI / link
Lukoff et al. [8]	2023	3-week within-person field deployment	606 survey; 46 field; 16 interviews	US Android YouTube users	SwitchTube search-first, recommendations or toggle	Yes: use/search logs; agency reported	Toggle improved agency; search-first did not minimize viewing time.	Toggle always last; app captured partial use; minimum-use requirement.	DOI
Lukoff et al. [78]	2021	Survey and participatory co-design	120 survey; 13 co-design	US YouTube users	Perceived agency across platform features	No: experience reports	Search/playlists felt agentic; recommendations/autoplay less so.	Attributions do not establish causal effects or reduced compulsivity.	DOI
Marchionini [77]	2006	Conceptual search framework	N/A	Information-seeking literature	Lookup, learning and investigation	No new data	Search includes open-ended exploration, not only bounded lookup.	Conceptual framework; not a compulsive-use study.	DOI
Guess et al. [6]	2023	See Table 2	See Table 2	See Table 2	Ranking comparison	See Table 2	Engagement effects summarized in Table 2.	Does not compare search with feeds.	DOI
Gauthier et al. [38]	2026	Preregistered 7-week field RCT	6,043 baseline; 4,965 post	US adult X users	Algorithmic versus chronological feed	Partial: content/follows logged; engagement reported	Algorithm effects asymmetric; see Table 2.	Not a direct search-versus-feed comparison.	DOI
Brady et al. [39]	2026	Registered 8-week feed RCT	2,000 randomized	Bluesky study participants	Engagement, chronological or diversified ranking	Yes: content/engagement logs	Content changed without significant own-engagement changes; see Table 2.	Not a direct search-versus-feed comparison.	DOI

Final synthesis

1 What science strongly supports

For specified platforms and interventions, engagement is causally responsive to interface and access changes. Strong examples include the large logged feed-ranking experiment and randomized app-limit/incentive study. This supports the tractability of behavior, not a universal clinical benefit from less use. [6, 3]

Self-reported use quantity is an inadequate replacement for logging. Behavioral quantity, subjective control and impairment must be measured as distinct constructs. [87, 88]

2 What science tentatively supports

Selective access, feature restriction and user-controlled modes can help align digital activity with intentions. Early HCI results justify prototyping a bounded retrieval interface, particularly for concrete tasks. The evidence does not yet establish broad retained utility, durability or superiority to blocking. [2, 1, 5]

3 What remains unknown

The crucial unknown is whether reliable external retrieval reduces **unwanted ecosystem engagement** without losing important information, interaction or valued discovery. The closest prototypes answer only parts of that question. Long-term adherence, cross-device displacement, clinical subgroup effects and the marginal value of AI summaries remain open. [5, 8, 2]

4 Evidence that challenges the product hypothesis

- Some checking begins from habit, boredom or affect regulation rather than a missing information item. [9, 10]
- Disabling external alerts did not reliably reduce logged use in a preregistered trial. [42]
- Search-first interaction did not simply minimize time. [8]
- Reduced target-app time can be displaced elsewhere; reduction can rebound. [3, 50]
- Abstinence does not produce uniform wellbeing improvements, and some behavioral improvements do not translate into problem-score changes. [47, 64]
- Extraction can omit meaning, while summarization can alter interpretation. [72, 75]

These findings constrain the mechanism rather than jointly disproving it. They suggest defining a narrower target population and distinguishing information retrieval from social participation and deliberately chosen leisure.

5 Most important research gaps

1. Direct randomization of external retrieval versus ordinary source access with matched content and separately measured cue exposure.
2. A credible extraction-versus-blocker trial with joint utility and unwanted-use endpoints.
3. Valid, low-burden episode measures distinguishing unplanned, unwanted and valued unexpected use.
4. Measurement of intermediary behavior, substitution and persistence beyond novelty.

5. Adequate study of users outside motivated student/technology samples and explicit evaluation of vulnerable groups.

These priorities follow the design and measurement limitations in the reviewed studies; they are proposed research directions, not established outcomes. [5, 2, 7, 87]

6 Ten highest value papers for the founding team to read

Paper	Why it deserves priority
Oulasvirta et al. 2012 [15]	Foundational checking and gateway mechanism.
Parry et al. 2021 [87]	Essential measurement warning before choosing metrics.
Allcott et al. 2022 [3]	Strong behavioral trial, explicit self-control model and substitution.
Lyngs et al. 2020 [1]	Direct selective-feed intervention with utility tradeoffs.
Lukoff et al. 2023 SwitchTube [8]	Close comparison of search, recommendations and user choice.
Orzikulova et al. 2023 FinerMe [2]	Closest app-versus-feature restriction comparison.
El Malki et al. 2026 Bonsai [5]	Closest natural-language intent and feed-curation prototype.
Guess et al. 2023 with corrections [6]	Large causal feed experiment; read both errata.
Dekker et al. 2025 Beyond the Buzz [42]	Strong challenge to a simple external-trigger explanation.
Monge Roffarello and De Russis 2023 [7]	Broad review of intervention effectiveness and methodological gaps.

7 Ten most important implications for product design

1. Start with verifiable information tasks and an explicit coverage boundary.
2. Make completion visible; require a choice for more content.
3. Preserve necessary replies, context and access to people.
4. Let users choose sources, schedules and whether they want exploration.
5. Separate ranking, supply limits, friction and summaries into testable components.
6. Avoid optimizing the intermediary for more time, queries or digest opens.
7. Make summary provenance and omissions inspectable.
8. Include overrides and evaluate frustration, access burden and abandonment.
9. Describe unwanted behavior without diagnosing or shaming users.
10. Optimize utility and unwanted engagement jointly, including substitutes.

These are proposed principles informed by the evidence; their individual support levels are specified in Section 13. They should not be marketed as ten independently proven treatment components.

8 Five highest priority experiments before the full product

1. **Retrieval fidelity and unmet needs:** establish accurate, timely coverage and identify utility that cannot be extracted.
2. **Matched app-entry/cue experiment:** separate source launch, cue exposure and content differences.
3. **Query-by-stopping-point factorial:** identify whether querying or bounded supply provides the benefit.
4. **Three-arm field trial:** usual access, credible blocking and extraction, with joint utility/noninferiority and unwanted-use/superiority tests.
5. **Delivery and persistence study:** measure recurring checking, substitution and social utility under schedules/adaptive delivery and after withdrawal.

Section 14 supplies proposed designs, durations, endpoints, power calculations and decision rules. The first study can restrict product scope; the others test efficacy and mechanism.

9 Proposed conceptual model

Source-app pathway: information need → app opening → cue/recommendation exposure → attentional capture or habitual continuation → unplanned consumption, some of which may be unwanted.

Extraction pathway: information need → explicit query or saved request → filtered content delivery → need satisfied → stopping or deliberately chosen continuation.

Arrow	Evidence grade for the specific relationship	Interpretation
Information need → source-app opening	Moderate descriptive, weak causal isolation	Goal-oriented phone activity occurs, but not every opening begins with such a need. [9]
Source-app opening → exposure to cues/recommendations	Strong within interfaces that display them	Interface experiments directly manipulate the exposure; this is not a claim about every app entry. [1, 6]
Exposure → attentional capture or continuation	Moderate	Notification, reward and autoplay experiments support particular links; settings and outcomes differ. [28, 24, 4]
Continuation → unplanned and unwanted use	Preliminary / mixed	Regret and goal conflict are context-dependent; continuation alone does not establish unwantedness. [35, 32]
Information need → explicit query or saved request	Preliminary	Prototype feasibility; unknown coverage of needs users cannot specify. [5, 70]
Query → correctly filtered delivery	Preliminary for cross-platform scope	Related systems demonstrate retrieval, not universal accuracy or completeness. [71, 70]
Delivery → valued need satisfied	Preliminary	Usability and user-control results are encouraging; comprehensive utility remains untested. [5, 72]
Need satisfied → natural stopping	Unsupported / insufficient direct evidence	This is the decisive proposed link; satisfaction could instead enable another query or return visit.

The model is not exclusively linear. Boredom, FOMO, social obligations, prior habit, task urgency and the surrounding situation can influence both pathways. The extraction interface can itself provide new cues and rewards. These are candidate moderators and feedback loops, not fully established causal paths. [10, 25, 30, 26]

10 Overall scientific assessment

Supported: interface choices can change engagement; habits and rewards contribute to some use; selective interventions can target particular behaviors. **Plausible extrapolation:** accurate delivery of a chosen item outside a feed can remove opportunities for unwanted continuation. **Not demonstrated:** that the complete intermediary preserves most utility, durably reduces compulsive behavior across the ecosystem, or outperforms blocking. [15, 24, 6, 2, 5]

The most informative favorable result would be a preregistered field trial showing less unwanted engagement, retained useful access and social functioning, and persistence without compensating use elsewhere, supported by a mechanism experiment that isolates cue exposure. The most informative unfavorable result would be high-fidelity extraction that clearly reduces cues and source entry yet fails to reduce unwanted engagement, or does so only by sacrificing valued utility. These are proposed evidential tests.

The concept should be pursued as a **conditional, falsifiable research hypothesis**. Its strongest initial scope is specific retrieval for people who repeatedly experience goal conflict, with transparent stopping and preserved actions. A claim to solve general social-media addiction would exceed the evidence reviewed here. [16, 32, 18]

Complete numbered bibliography

Numbers correspond to inline citations. Study design, sample, population, findings and limitations appear in the seven evidence tables or the instrument appraisal; additional study details are included below for sources outside those tables. NR means not verified in the accessible original record. Conceptual and methodological publications have no new efficacy sample. DOI links lead to the publication record; original-source links provide publisher, PubMed or author/institutional access.

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